

Data Flow Diagram

Date	2 October 2026
Team ID	SWTID-2026-5625
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	

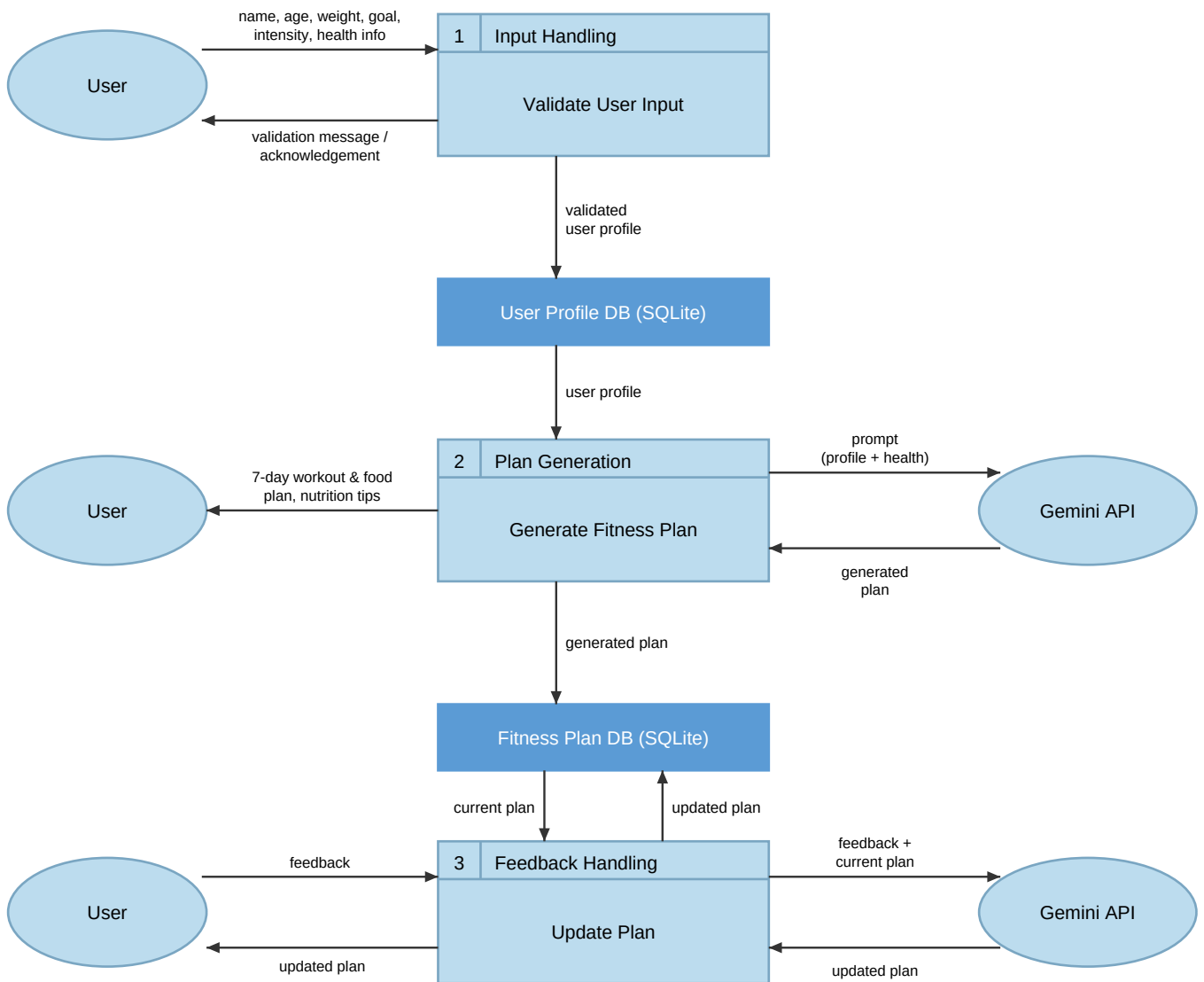
Data Flow Diagram

The Data Flow Diagram (DFD) below shows how data moves through FitBuddy – between the external entities (User and Google Gemini API), the processes (input handling, plan generation, feedback handling) and the data stores (User Profile DB and Fitness Plan DB).

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person or system outside FitBuddy that sends or receives data (e.g., User, Gemini API).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate User Input, Generate Fitness Plan).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., User Profile DB, Fitness Plan DB).
Labeled arrow	Data Flow	The movement of data between entities, processes and data stores, labeled with the data passed.

FitBuddy – Data Flow Diagram



Process Summary

No.	Process	What it does
1	Input Handling	Receives name, age, weight, fitness goal, workout intensity and optional health information from the user via the FastAPI form, validates it and saves the profile.
2	Plan Generation	Builds a prompt from the stored profile, sends it to Gemini, receives the 7-day workout and food plan with nutrition and recovery tips, stores it and shows it to the user.
3	Feedback Handling	Takes the user's feedback and the current plan, asks Gemini for a revised plan, updates the Fitness Plan DB and returns the updated plan.

Team Members and Roles

1. Sharmila D – Team Lead

- Managed the overall project development.
- Handled project pre-requisites and topic links.
- Worked on the core functionalities of FitBuddy.
- Coordinated the team members and project activities.
- Prepared the application for local deployment.

2. Bhavana A – Backend Developer

- Worked on the project workflow.
- Developed and handled the FastAPI backend.
- Worked on routing and user-input handling.
- Tested the application during local deployment.

3. Rajashree K – AI/GenAI Developer

- Worked on Gemini AI integration.
- Researched the suitable Gemini model for the project.
- Worked on the main application logic.
- Contributed to the conclusion and AI-related functionality.

4. Siva Sundari V – Frontend/UI Developer

- Worked on the application architecture.
- Designed the user interface.
- Developed the frontend components.
- Focused on creating a simple and user-friendly interface.

5. Praveena K – Development and Template Developer

- Worked on the development environment setup.
- Developed dynamic templates using FastAPI and Jinja2.
- Worked on connecting the frontend templates with the backend.
- Supported application development and testing.